

LINEAR REGRESSION

Module 3: Predicting the Future

1. THE PREDICTION PROBLEM

Linear Regression is the mathematical process of finding the “best-fitting” straight line through a cloud of data points. It asks: *If I know X, what is the most likely value of Y?*

- **The Linear Equation:**

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Y : The dependent variable (what we want to predict).
- X : The independent variable (what we know).
- β_0 : The Intercept (where the line starts on the Y-axis).
- β_1 : The Slope (how much Y changes for every 1 unit of X).
- ϵ : The “Error Term” (the distance between reality and our prediction).

2. ORDINARY LEAST SQUARES (OLS)

How do we define the “best” line? We minimize the total distance between every real data point and our line. Because some points are above the line (positive error) and some are below (negative error), we **square** the distances so they don’t cancel each other out.

- **The Goal:** Minimize the Sum of Squared Residuals (SSR):

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **The Result:** The line that produces the absolute minimum SSR is the most mathematically “honest” prediction possible.

3. THE 3D LOSS LANDSCAPE

In modern AI, we don't just solve one equation; we use **Gradient Descent**. Imagine a 3D bowl where the bottom of the bowl represents the lowest possible error (the best line).

- **The Sphere:** Think of your algorithm as a ball dropped into this 3D bowl.
- **The Descent:** The ball rolls downward, constantly adjusting the Slope (β_1) and Intercept (β_0) until it rests at the exact bottom (the "Global Minimum"). This is exactly how deep learning models "learn" to improve their accuracy over time.

4. ASSUMPTIONS OF LINEARITY

A model is only as good as the data it's fed. Linear regression assumes:

1. **Linearity:** The relationship is a straight line, not a curve.
2. **Independence:** One data point doesn't influence the next.
3. **Homoscedasticity:** The error (spread) is consistent across all values of X.

If these assumptions are violated, your prediction line will be fundamentally flawed.

5. BEYOND PREDICTION

Once you master the straight line, you have unlocked the basic logic of every AI in existence. But what happens when the relationship is not a straight line? What happens when you need to predict a category rather than a number? That is where we bend the line into an S-curve.